

Joel Jacob

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Portfolio

Education

Carnegie Mellon University – BS in Mechanical Engineering

Expected May 2028

Skills

Design & Analysis: SolidWorks, Onshape, ANSYS (Discovery, Fluent), Tinkercad, MS Office

Controls & Programming: Python, MATLAB, Java, C++, Arduino, Raspberry Pi, Git

Manufacturing & Fabrication: 3D Printing, Machining, Laser Cutting, Soldering, Mechanical Assembly

Experience

Research Assistant, Carnegie Mellon University – Pittsburgh, PA

Sept 2025 – Present

- Contributed to the development of a wearable assistive device as a member of the MetaMobility Lab at CMU by integrating sensing and deep learning to improve human mobility support
- Redesigned hip exoskeleton hardware for improved user fit by applying human biomechanics principles, leveraging SolidWorks and 3D printing to achieve precise motor-joint alignment and a 30% weight reduction
- Integrated laboratory devices using Python to automate data collection and processing, improving experimental workflow and data quality

Projects

NASA Lunabotics Challenge

Nov 2025 – Present

- Designed and fabricated excavation subsystem components as a member of CMU Moon Miners, winning the best presentation by a rookie team at the 2026 NASA Lunabotics Competition
- Owned the dual-channel bucket excavator design through design reviews, 3D printed prototyping, and machined final version that outperformed the baseline in terms of regolith throughput
- Minimized weight through structural FEA in ANSYS Discovery while maintaining safety factors

National Havoc Robot League

Jan 2025 – Present

- Founded a new robot project under CMU Combat Robotics, coordinating a team to design, develop, and manufacture a 3 lb undercutter combat robot to compete in the National Havoc Robot League
- Designed 10+ custom robot components on Onshape, including a TPU chassis, urethane wheels, and AR500 steel spinner weapon, accelerating iteration before 3D printing, rubber casting, and machining
- Optimized ideal motor Kv and weapon moments of inertia using MATLAB analysis to maximize impact effectiveness while retaining stable rotational performance

FIRST Robotics Competition

Oct 2021 – June 2024

- Oversaw the 12-member control department as a lead of FRC Team 5123 to program Java command-based code and network a motor controller system to win the 2023 FRC NYC Regional Championship
- Improved autonomous driving accuracy and reduced steady-state error by 65% by implementing vision-based localization with Raspberry Pi and tuning PID gains